Visual Modality

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Visual modality includes different media types: Text, static and dynamic images and 3D graphics.

Part One – Text modality

Text is a set of symbols that, when combined according to grammatical rules, convey meaning.

1. Communication Dimension

In the context of human–machine interaction, communication through text is straightforward: it involves the correspondence between **numeric data (codes)** and **symbols**.

2. Interpretation

It functions as a **symbolic communication system**, (~ sign language). Each symbol carries a **semantic value**, which allows meaning to emerge from structured combinations.

3. Utility

Within multimedia applications, text is often integrated into scenes to:

- ➔ Provide additional information and details;
- ➔ Clarify or translate conversations;
- ➔ Describe contexts and situations.

4. Implementation

Text is the **simplest communication channel** and the most accessible perceptual dimension. It offers advantages such as: low storage cost, efficient encoding, and ease of manipulation.

5. Advanced Tools

Advanced text processing relies on computational tools such as:

- ➔ Lexical analyzers;
- ➔ Syntax analyzers;
- ➔ Semantic parsers;
- ➔ Generative AI Tools

Part Two – Vision

1. Definition

Vision in multimedia refers to the digital representation of visual information, allowing users to perceive and interpret images or graphics through the sense of sight.

2. Digital Representation (Computer Model)

In pixel-based model (Raster/Bitmap), each pixel is associated with a value in a model, defined by:

- ➔ **Depth (Bit Depth) –** Number of bits used to represent each pixel (*8-bit grayscale, 24-bit RGB*).
- ➔ **Color definition** – Grayscale, RGB, CMYK, or other color models.
- ➔ **Image Resolution –** The number of pixels sampled; higher resolution → greater detail.
- ➔ **Render (display) definition –** Maximum number of pixels displayed by a device.

Vector images are defined by mathematical descriptions of shapes, lines, and curves, allowing infinite scalability without loss of quality.

A **color model** is a mathematical and conceptual framework used to represent colors as ordered sets of numerical values: Each color is encoded using one or more components (*e.g., red, green, and blue*), allowing precise storage, manipulation, and reproduction of visual information.

Available on zenodo – DOI : 10.5281/zenodo.17251328

3. Vector images

Vector images are defined **mathematically** using *algorithms and primitives* (points, lines, curves, polygons, transformations). They are *resolution-independent* because the description can be scaled infinitely without quality loss.

Screens, however, are made of **physical pixels**. No matter the original format, the display device can only show **bitmaps** (a rasterized grid of pixels).

When a vector image is displayed on screen, it undergoes a process called rasterization: the mathematical description is converted into a bitmap adapted to the screen resolution.
Printing devices work similarly: even if you send vector graphics to the printer, they are rasterized internally according to the printer's DPI.

3.1. How is vector image information actually encoded in a file?

Unlike bitmap images, which are stored **pixel by pixel**, a vector image is stored as **mathematical instructions** describing shapes, colors, and the arrangement of elements.

1. **Graphic Primitives (the "building blocks")**
   - ➔ **Points** (x, y coordinates);
   - ➔ **Simplices** (Vertices, Edges, Facets and Faces);
   - ➔ **Curves and Surfaces** (Bezier curves, splines, arcs);
   - ➔ **Text** (vector fonts, where each character is a shape).
2. **Visual Attributes:**
   Stroke color, Stroke width (line thickness), Fill color, Gradients, patterns, transparency…
3. **Geometric Transformations:**
   Translation (move), Rotation, Scaling (zoom in/out)…
4. **Hierarchical structure:**
   Objects are organized into layers or groups.

4. Types of Visual Media

Still images – Static visual representations, such as photos or graphics.
Moving images (animated) – Dynamic sequences: Videos, Computer Animations…

5. Utility

Visual media provide rich information and context within multimedia applications, including:

- ➔ Representation of real or simulated environments;
- ➔ Communication of abstract concepts through diagrams and illustrations;
- ➔ Enhancement of user engagement through animations and visual storytelling.

6. Implementation

Vision is one of the most **data-intensive** communication channels. Storage and processing requirements depend on image type:

- ➔ **Raster images** require significant storage and compression (JPEG, PNG).
- ➔ **Vector images** require less storage and allow easy transformations.
- ➔ **Video and animations** involve continuous image sequences, demanding high bandwidth and efficient compression (MPEG, MP4, AVI).

7. Advanced Tools

Working with visual media involves a variety of techniques and tools, such as:

- ➔ Image Editing (*Photoshop, GIMP*).
- ➔ Vector Design (*Illustrator, Inkscape*).
- ➔ Video Editing and post-production (*Premiere, After Effects*).
- ➔ Rendering engines and animation software (*Blender, Unity*).